Lab 8: Power Analysis: Part 2

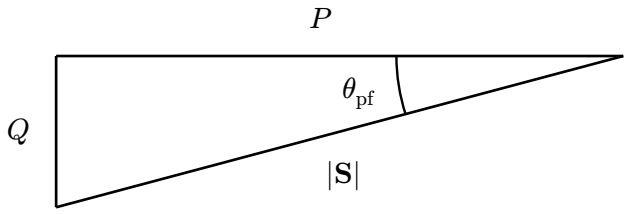
ECE2200L Section 04, Fall 2026

Matthew Ponciano, Eli Pantoja

October 12, 2026 - October 26, 2026

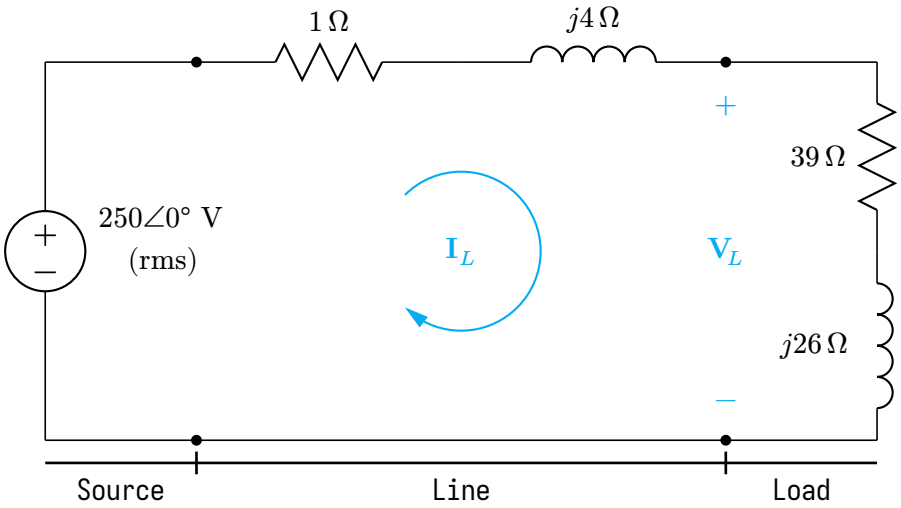
No 1.

Step 1. The power triangle that characterizes the impedance in a circuit is shown below. In its simplest form, what does the impedance consist of?



The impedance consists of some (positive) resistance and some (negative) reactance. Thus, a simplest equivalent model is a resistor and capacitor in series, because the circuit absorbs real power and has negative reactive power.

No 2.



Step 1. Find the phasor voltage $\mathbf{V}_L$ and current $\mathbf{I}_L$ for the load.

$$\begin{aligned} Z_{\text{eq}} &= Z_{\text{line}} + Z_{\text{load}} \\ &= (1 + j4) + (39 + j26) \\ &= 40 + j30\,\Omega \end{aligned}$$

$$\mathbf{V}_L = (250\angle 0°)\left(\frac{Z_{\text{load}}}{Z_{\text{load}} + Z_{\text{line}}}\right)$$

$$\boxed{\mathbf{V}_L = 234 - j13\,\text{V} = 234.36\angle -3.18°\ \text{V}}$$

$$\mathbf{I}_L = \frac{\mathbf{V}_L}{Z_{\text{load}}}$$

$$\boxed{\mathbf{I}_L = 5\angle -36.87°\ \text{A} = 4 - j3\,\text{A}}$$

Step 2. Show that the power in the circuit balances.

$$\sum \mathbf{S} = \mathbf{S}_s + \mathbf{S}_{\text{line}} + \mathbf{S}_{\text{load}}$$

$$\mathbf{S}_s = -\mathbf{V}_s \mathbf{I}_L^* = -1000 - j750\ \text{VA}$$
$$\mathbf{S}_{\text{line}} = |\mathbf{I}_L|^2 Z_{\text{line}} = 25 + j100\ \text{VA}$$
$$\mathbf{S}_{\text{load}} = |\mathbf{I}_L|^2 Z_{\text{load}} = 975 + j650\ \text{VA}$$

the source is delivering power, so its power is negative:

$$\sum \mathbf{S} = -(1000 + j750) + (25 + j100) + (975 + j650)$$

$$\boxed{\sum \mathbf{S} = 0 \quad \checkmark}$$

Step 3. Use PSpice to simulate the circuit to confirm your results in Steps 1 and 2.

Assuming $f = 60$ Hz,

$$X_L = \omega L \quad \Rightarrow \quad L = \frac{X_L}{2\pi f} \quad \Rightarrow \quad L_{\text{line}} = \frac{4}{2\pi 60},\quad L_{\text{load}} = \frac{26}{2\pi 60}$$

View simulation code

```
V_L: 234.00 -13.00j V = 234.36 ∠ -3.18° V
I_L: 4.00 -3.00j A = 5.00 ∠ -36.87° A
S_s (absorbed): -1000.00 -750.00j VA
S_line: 25.00 +100.00j VA
S_load: 975.00 +650.00j VA
Power balance: 0.00 +0.00j VA
```

No 3.

Step 1. A load requires a voltage of $240\,\text{V}$ (rms), absorbs $8\,\text{kW}$ and has a power factor of 0.8 lagging. Find the complex power of the load.

$$\mathbf{V} = 240\angle 0°\ (\text{rms}),\quad P = 8000\,\text{W},\quad \text{pf} = 0.8\ \text{lagging}$$

$$\varphi = \cos^{-1}(\text{pf}) \Rightarrow \varphi = 36.87°$$
$$Q = 8000\tan(\varphi) \Rightarrow Q = 6000\ \text{VAR}$$

$$\Rightarrow \boxed{\mathbf{S} = 8000 + j6000\ \text{VA}}$$

Step 2. Find the load impedance.

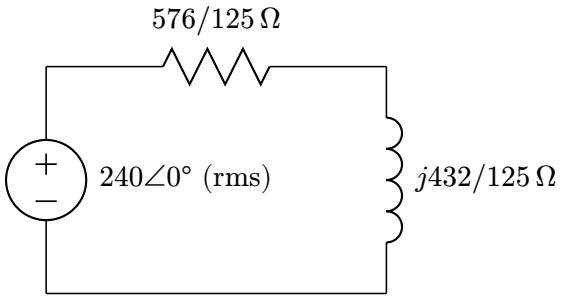
$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = \mathbf{V}\left(\frac{\mathbf{V}}{Z}\right)^* = \mathbf{V}\frac{\mathbf{V}^*}{Z^*} = \frac{|\mathbf{V}|^2}{Z^*}$$

$$Z^* = \frac{|\mathbf{V}|^2}{\mathbf{S}}$$

$$Z = \frac{|\mathbf{V}|^2}{\mathbf{S}^*}$$

$$\boxed{Z = \frac{576}{125} + j\frac{432}{125}\,\Omega = 5.76\angle 36.87^\circ\,\Omega}$$

Step 3. Estimate the circuit configuration.



Step 4. Simulate your findings and confirm the results in Steps 1-3.

assuming $f = 60$ Hz,

$$L = \frac{432/125}{2\pi 60} \approx 9.167\,\text{mH}$$

View simulation code

```
Inductance = 9.167 mH
Voltage = 240.00 V (rms)
Current = 33.333 -25.000j A
|Current| = 41.667 A (rms)
Current phase relative to voltage = -36.87 degrees
Complex power = 8000.00 +6000.00j VA
Impedance = 4.608 +3.456j ohm
Power factor = 0.800
```